

NING ZHANG

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Education

DPhil at Department of Statistics, University of Oxford

Fundamental Methodologies for Machine Learning

Jan 2023 – Current

Oxford, England

MASc in ECE, University of British Columbia

GPA: 93.8/100; Thesis topic: statistical estimation for graph alignment

Sep 2019 – Apr 2022

Vancouver, Canada

BSc in Physics (*Honor*), Nankai University

GPA: 90.8/100; Ranking: 3/92; Thesis topic: statistical physics inspired image analysis

Sep 2015 – Jun 2019

Tianjin, China

Research Interest

graph theory, spectral methods, statistical machine learning, applied probability and statistics, optimization, algorithms, statistical physics

Publications

1. **Ning Zhang**, Henry Kenlay, Li Zhang, Mihai Cucuringu, and Xiaowen Dong. On the stability of graph convolutional neural networks: A probabilistic perspective, 2025
2. **Ning Zhang**, Xiaowen Dong, and Mihai Cucuringu. Spectral clustering for directed graphs via likelihood estimation on stochastic block models. 2025
3. Ziao Wang, **Ning Zhang**, Weina Wang, and Lele Wang. On the feasible region of efficient algorithms for attributed graph alignment. In *IEEE Transactions on Information Theory*, 2024
4. **Ning Zhang**, Ziao Wang, Weina Wang, and Lele Wang. Attributed graph alignment. *IEEE Transactions on Information Theory*, 2024
5. **Ning Zhang**, Susan Francis, Rayaz A Malik, and Xin Chen. A spatially constrained deep convolutional neural network for nerve fiber segmentation in corneal confocal microscopic images using inaccurate annotations. In *2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI)*, pages 456–460. IEEE, 2020

Awards

- 2023-2027 DPhil Scholarship from EPSRC and IBM
- 2021 IEEE North American School of Information Theory Best Poster Award
- 2020 Honourable Mention in Graph Attack and Defence Track of KDD Cup (rank: 14/106)
- 2019 Outstanding Graduate in Nankai University (3%)
- 2017 Gong Neng Award (10%)
- 2016 & 2018 The Second & First Prize Scholarship for Outstanding Student (10%, 5%)
- 2015-2019 Poling Scholarship (National Top-talent Students Training Program)

Working Experience

Microsoft Research Lab - Asia

Research Intern, Advisor: Haoyu Dong

Oct 2022 – Jan 2023

Beijing, China

Statistical methods for robust computation on Large Language Models.

Teaching

- MT 2024, HT 2025 Graph Signal Processing
- MT 2024 Graph Representation Learning
- MT 2023, 2024 Probability and Statistics for Network Analysis
- Fall 2020, 2021 Stochastic Signals and Systems (STAT321)
- Spring 2020 Electrical Engineering Design Studio I (ELEC291)

Talks and Activities

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| 2024 | On the Stability of Graph Spectral Filters: from a Probabilistic Perspective
<i>Graph Signal Processing Workshop, Delft, Netherlands, June 2024</i> |
| 2023 | Maximum Likelihood Estimation on Stochastic Blockmodels for Directed Graph Clustering 
<i>International Conference on Complex Networks, Menton, France, Dec, 2023</i> |
| 2021 | Attributed graph alignment 
<i>IEEE International Symposium on Information Theory (ISIT), June 2021</i> |
| 2020 | A Spatially Constrained Deep Convolutional Neural Network for Nerve Fiber Segmentation 
<i>IEEE International Symposium on Biomedical Imaging (ISBI), April 2020</i> |

Relevant Skills

Language: Chinese (native), English (fluent)

Coding languages: Python, MATLAB, Mathematica, C++

Technologies/Frameworks: Linux, Github